Causal Meta-Transfer Learning for Cross-Domain Few-Shot Hyperspectral Image Classification

Yuhu Cheng, *Member, IEEE*, Wei Zhang, Haoyu Wang, *Member, IEEE*, and Xuesong Wang, *Member, IEEE*

***Abstract*— Few-shot hyperspectral image (HSI) classification poses challenges due to sample selection bias in few-shot scenarios, potentially leading to incorrect statistical associations between noncausal factors and category semantics. To address these challenges, an original HSI is treated as a mixture comprising causal and noncausal factors. By integrating the causal learning, meta-learning, and transfer learning, a cross-domain few-shot HSI classification method based on causal meta-transfer learning (CMTL) is developed. First, a mask Transformer is implemented to identify noncausal factors unrelated to categories. Second, an independent causal constraint is applied to separate the causal and noncausal factors and enhance the inclusion of pure and independent causal factors in the features. Finally, the meta-transfer learning is leveraged to enable the classification model to extract causal factors highly correlated with category semantics from data, facilitating the cross-domain knowledge transfer. Meanwhile, a causal association module (CAM) is employed to maximize the mutual information between causal factors and category predictions, thereby ensuring a strong causal association between causal factors and classification tasks. Experimental results show that the CMTL achieves competitive performance in cross-domain few-shot HSI classification tasks.**

***Index Terms*— Causal learning, cross-domain knowledge transfer, few-shot classification, hyperspectral image (HSI), meta-learning.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is an advanced information processing method that integrates artificial intelligence and satellite remote sensing technology [1], [2]. By leveraging the rich spatial and spectral information contained in HSIs, HSI classification enables intelligent recognition of pixel point categories, playing a critical role in ensuring production safety, preventing natural disasters, and promoting mining industry [3], [4], [5].

In recent years, deep learning-based methods for HSI classification have been favored by researchers. Among them, the most widely used methods can be divided into three categories: convolutional neural network (CNN) [6], [7], [8], graph neural network (GNN) [9], [10], [11], and Transformer [12]. As the most widely used backbone network at present, CNN demonstrates powerful feature extraction capabilities in HSI classification. Zhang et al. [13] proposed a CNN-based structurally optimized transmission network that enhances the robustness of feature extraction and classification by capturing the spectral and spatial structure information of multisource remote sensing data. GNN has excellent spatial structure capture capability, thus effectively dealing with HSI with irregular spatial distribution and realizing the identification feature extraction by making full use of spatial context information. Ding et al. [14] proposed a novel semisupervised locality-preserving dense GNN with autoregressive moving average filters and context-aware learning for HSI classification, which captures the global graph structure to improve the robustness of classification model for noisy information. The Transformer, with its powerful capability to capture long-range dependencies, delivers excellent performance in HSI classification tasks. Yang et al. [15] proposed a novel HSI Transformer, which can capture subtle spectral differences and convey local spatial context information by embedding convolution operations in Transformer. He et al. [16] proposed HSI-BERT with global sensitivity field, where HSI stands for hyperspectral image, and BERT stands for bidirectional encoder representations from transformers. The HSI-BERT can fully mine spatial-spectral features by capturing global dependencies among pixels and improve the generalization performance of the HSI classification model.

However, in HSI classification studies, few-shot scenarios often arise due to the time-consuming and laborious labeling process of HSIs, which can lead to limitations in the performance of deep learning methods that heavily rely on data. Consequently, minimizing the dependence of classification models on target HSI labeling information is crucial for achieving accurate classification in few-shot scenarios.

In recent years, the role of transfer learning in HSI classification tasks has attracted increasing attention from researchers. In particular, it has shown promise in scenarios with scarce annotation data in the target domain. Transfer learning leverages the data and knowledge from a labeled source domain to assist in accomplishing few-shot classification tasks in similar but distinct target domains [17]. Transfer learning-based few-shot classification methods are generally categorized into two types: fine-tuning-based and domain-adaptation-based. Fine-tuning-based methods initially train the classification model on a source domain with an adequate number of training

Manuscript received 21 July 2023; accepted 17 August 2023. Date of publication 28 August 2023; date of current version 8 September 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62176259 and Grant 62373364, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20221116, in part by the Excellent Post-Doctorate Program of Jiangsu Province under Grant 2022ZB530, and in part by the Key Research and Development Program of Jiangsu Province under Grant BE2022095. *(Corresponding author: Xuesong Wang.)*

The authors are with the Engineering Research Center of Intelligent Control for Underground Space, Ministry of Education, and the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou 221116, China (e-mail: chengyuhu@163.com; zweicumt@163.com; wanghaoyucumt@163.com; wangxuesongcumt@163.com).

Digital Object Identifier 10.1109/TGRS.2023.3309055

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

samples and then transfer the model to the target domain for fine-tuning with limited number of samples. Zhang et al. [18] pretrained some classification models on various hyperspectral and natural light images and subsequently fine-tuned them on the target domain, achieving cross-sensor, cross-modal, and cross-scene classification of few-shot HSIs. Hang et al. [19] proposed a fine-tuning strategy that employs cross-modal data for model training and fine-tunes the classifier with a small number of training samples to facilitate the cross-domain knowledge transfer. Domain-adaptation-based methods reduce the distribution difference between two domains and then use the adapted data to complete the target-domain classification. Qu et al. [20] bridged the distribution difference by mapping HSI from source and target domains into a shared subspace based on their respective physical features and accomplished the cross-scene few-shot classification task by merely using source-domain labeled samples and a small number of samples from the target domain. Zhang et al. [21] used a graph structure to represent topological relationships of HSIs in two domains and aligned them using graph optimal transmission, while also aligning the topological and feature distributions of the two domains. Wang et al. [22] used the graph embedding structure as a category discriminative transfer feature learning term and cross-domain alignment term and achieved category discriminative transfer by aligning the local and global geometric structures across domains. Zhang et al. [23] extended the domain adaptation of single source data to the collaborative domain adaptation of multisource data, constructed a modal-related domain-adaptation network, and utilized complementary classifiers to aggregate all discriminative information captured from different modalities. The transfer of features from the source to target domains is achieved by aligning the local and global geometric structures across domains. Such methods do not require the train and test sets to obey the same distribution, but still require the two domains to have the same categories of land cover. However, differences in spatial regions at the time of acquisition may result in inconsistent feature categories across different HSIs, which will hamper the knowledge transfer between domains.

In the task of HSI classification, the meta-learning simulates the scenario of few-shot classification of heterogeneous land cover in the target domain by constructing a few-shot scenario task and summarizes general few-shot classification knowledge from it. Therefore, the few-shot classification method based on meta-learning could address the issue of inconsistent categories between the source and target domains [24], [25]. Bai et al. [26] used meta-learning as a training mechanism to combine subspace techniques and feature transformations to achieve few-shot HSI classification by distancing different category subspaces. Liu et al. [27] proposed a deep few-shot learning (DFSL) that used the rich source-domain information to learn the category metric space and classify unlabeled data of test set. Gao et al. [28] introduced a relational network [29] into the few-shot HSI classification task by learning the metric space and metric network simultaneously. Zuo et al. [30] proposed an edge labeling GNN for few-shot HSI classification, which utilizes a GNN to capture the intracategory and intercategory correlations between samples.

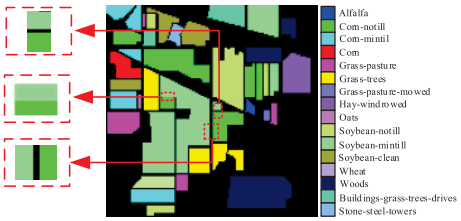


Fig. 1. False statistical association.

Few-shot learning requires the source and target domains to have the same data distribution. However, the distributions of HSIs in these two domains are different, which poses challenges for the application of few-shot learning in HSI classification. The cross-domain few-shot learning aims to solve the problem of knowledge transfer. This type of method combines the meta-learning and transfer learning into a unified framework. It aims to use meta-learning to summarize the meta-knowledge suitable for few-shot classification tasks and uses transfer learning to align the two-domain data distribution, thereby promoting cross-domain knowledge transfer. Wang et al. [31] proposed a graph meta-transfer network (GMTN), which achieved spatial–spectral feature extraction by combining the local information extraction ability of graph learning and attention mechanism. Li et al. [32] proposed a novel deep cross-domain few-shot learning (DCFSL) method that combines a conditional adversarial domain-adaptation strategy with meta-learning to accomplish HSI classification under few-shot conditions. Zhang et al. [33] proposed the graph information aggregation cross-domain few-shot learning (Gia-CFSL), which merges the few-shot learning and domain adaptation via the graph information aggregation. It captures nonlocal dependencies and minimizes feature distribution and graph topology differences. Wang et al. [34] used a pretrained model on natural image data to fine-tune few-shot learning to achieve knowledge transfer between heterogeneous datasets of natural images and HSIs. In addition, Wang et al. [35] introduced the attention-like and asymmetric adversarial training in the framework of few-shot learning to achieve more discriminative domain-invariant feature extraction.

The existing meta-transfer learning methods have achieved impressive performance on few-shot HSI classification tasks. However, they often neglect the possibility that sample selection bias may result in spurious statistical associations between noncausal factors and sample categories, which will subsequently affect the generalization performance of model. For instance, in the Indian Pines (IP) dataset, the "Corn-notill" and "Soybean-mintil" categories are frequently located close to each other and exhibit high statistical dependence, as shown in Fig. 1. If the support sample for "Corn-notill" contains information about "Soybean-mintil," the model may incorrectly associate the two categories, leading to a false statistical association that is difficult to generalize [36].

By identifying causal factors related to category semantics, such as pixels, which contain abundant class discriminability information, the causal learning can enhance a model's generalizability capability while mitigating the negative impacts of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

noncausal factors on the model, including anomalous spectral information and heterogeneous pixel information. The existing studies [37], [38] emphasize the significance of selecting causal factors that demonstrate three key characteristics: 1) independence from noncausal factors; 2) joint independence among causal factors; and 3) strong correlation with category semantics. We proposed a novel cross-domain few-shot HSI classification method based on causal meta-transfer learning (CMTL). The main contributions of this article include the following.

1) A CMTL framework is proposed, which helps the model learn transferable causal knowledge by using the scenario training mechanism and removes the false statistical association between noncausal factors and category semantics.
2) A causal intervention module (CIM) based on mask Transformer is constructed, which utilizes multiview information from spectral and spatial for causal mask learning, intervenes in noncausal factors based on causal masks, and leverages independent causal constraints to separate causal and noncausal factors effectively.
3) A causal association module (CAM) is constructed, which uses nonlinear mapping to capture the marginal distribution and joint probability distribution between features and category predictions and ensures strong causal correlation between features and category semantics through mutual information optimization.

II. CMTL for HSI Classification

A. Method Overview

The CMTL primarily consists of five components: the dimension unification module (DUM), feature extraction module, CIM, CAM, and meta-transfer learning. The DUM is used to adjust the feature dimensions of the source- and target-domain HSIs. The feature extraction module is responsible for extracting discriminatory features. The CIM generates counterfactual HSIs and separates causal factors from noncausal factors through independent causal constraints to ensure that the feature representation merely contains pure causal factors. The meta-transfer learning utilizes the adversarial domain-adaptation module (ADAM) to effectively diminish the distribution difference between the source and target domains. It inducts transferable knowledge for few-shot classification by implementing prototype matching of both support and query features within the feature space. The CAM ensures a strong correlation between causal factors and category semantics by maximizing the mutual information between the feature and category prediction.

B. Framework of CMTL for HSI Classification

The training process of CMTL is illustrated in Fig. 2, which primarily consists of two parts: the extraction of causal factors and meta-transfer learning.

The first part is extraction of causal factors. First, the source-domain and target-domain HSIs are initially fed into the dimensionality communication module. This process yields unified dimensional training data. Subsequently, training data from both domains is partitioned into support sets and query sets, which leads to the construction of few-shot scenario tasks. Then, in few-shot scenario tasks, the support sets and query sets are input into the mask Transformer, resulting in a causal mask that can segment each HSI into causal and noncausal regions. Subsequently, based on this causal mask, pixels within the noncausal regions are substituted with randomly chosen heterogeneous pixels. This procedure perturbs noncausal factors to derive the counterfactual HSI. Following this, the original and counterfactual HSIs are fed into the feature extraction module, and an independent causal constraint is applied. The final outputs of this stage include support and query features, both of which contain pure and independent causal factors.

The second part is meta-transfer learning. First, the support and query features from the two domains are input into the ADAM to alleviate the distribution difference between domains. Subsequently, the support features are employed to compute category prototypes, while category predictions for the query features are obtained through prototype matching. Ultimately, the query features and its associated category predictions are fed into the CAM. The strong causal association between the feature and category prediction is ensured by maximizing their mutual information.

C. DUM Based on PCA

Due to differences in collection devices, source- and target-domain HSIs often have different spectral band numbers. To obtain data from both domains with the same dimensions, it is necessary to standardize the number of spectral bands of the HSI. We choose principal component analysis (PCA) [39] to unify the feature dimensions of the source and target domains, as it has the following advantages. On the one hand, PCA is straightforward and does not heavily rely on specific parameter assumptions. On the other hand, PCA can maximize the preservation of original HSI information, thereby reducing the loss of important information caused by dimensionality reduction.

D. Causal Intervention Based on Mask Transformer

Due to the complex spatial distribution of land cover in HSIs, heterogeneous land cover information frequently causes significant information interference, particularly for HSIs located in spatial boundary regions. In addition, due to the existence of phenomena, such as “different objects have the same spectra” and “the same objects have different spectra,” some HSIs often contain spectral noise. The heterogeneous land cover information and spectral noise can be regarded as noncausal factors unrelated to sample category. Because noncausal factors are often tightly coupled with causal factors in HSIs, it is difficult to separate causal factors.

Causal intervention [40] facilitates the separation of causal factors from data by perturbing noncausal factors. Nevertheless, the spatial information that corresponds to the heterogeneous land cover and spectral anomalies in HSIs is often

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

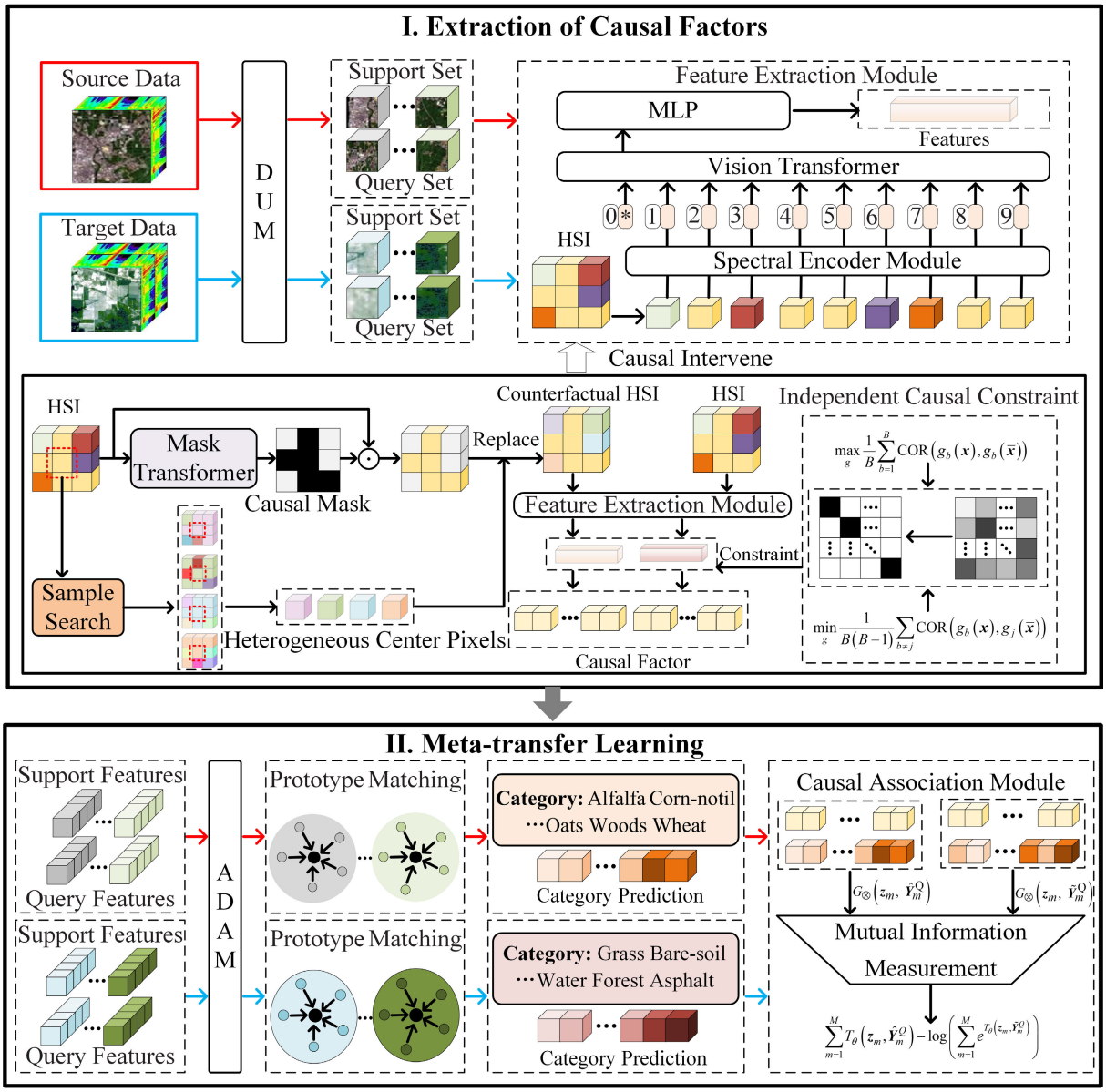


Fig. 2. Framework of CMTL.

indeterminate, complicating the process of locating and perturbing these factors. To combat this challenge, we proposed a CIM based on mask Transformer. The inference process of the causal mask for individual target pixels is shown in Fig. 3. During the training process, it is worth noting that iterations are not performed on individual pixels but are conducted in batches, with each batch comprising all the target pixels within the HSI patches. The primary workflow consists of the following steps. First, the differences between causal and noncausal factors are captured to obtain the mask vector for different land cover pixels in HSIs. Second, the causal mask matrix is derived based on the mask vector to identify noncausal factors within the spatial neighborhood. Finally, based on the causal mask matrix, causal interventions are performed on the HSIs to obtain the counterfactual HSIs. By maximizing the correlation between the counterfactual HSIs and original HSIs in the feature space, the causal factors that remain invariant under causal interventions are captured. The specific details of this process are as follows:

1) Causal Mask: To obtain the causal mask and identify the regions containing noncausal factors, we employ the mask Transformer, which primarily comprises spatial perception, spectral feature extraction, and causal mask learning.

The first part is spatial perception. The spatial neighborhood information of the target pixels (noncentral pixels in HSI patch) and center pixel (pixel to be classified) is aggregated by using the average pooling to obtain the target local pixel set and center local pixel set.

The second part is spectral feature extraction. The spatial perceptual set that contains center pixel, target pixel, average pooled center local pixel set, and average pooled target local pixel set is input into the spectral encoder module to obtain the spatial perceptual features set. The spatial perception feature set of the ith target pixel in HSI patch can be denoted as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

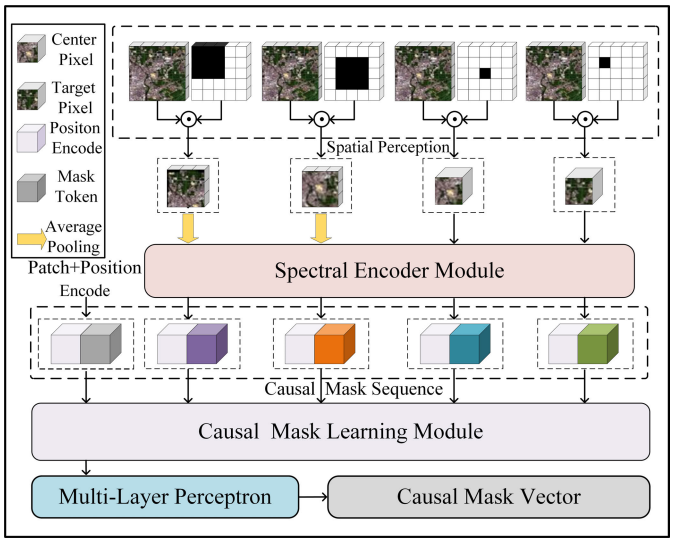


Fig. 3. Framework of mask Transformer.

follows:

$$\tilde{z}_i^{\text{spa}} = \left\{\tilde{z}_i^{\text{cen}}, \tilde{z}_i, \hat{z}_i^{\text{cen}}, \hat{z}_i\right\} \tag{1}$$

where $\tilde{z}_i^{\text{spa}}$ represents spatial perception feature set, $\tilde{z}_i^{\text{cen}}$ represents the spectral feature of the central pixel, $\tilde{z}_i$ denotes the spectral feature of the target pixel, $\hat{z}_i^{\text{cen}}$ signifies the spatial local feature of the central pixel, and $\hat{z}_i$ corresponds to the spatial local feature of the target pixel.

The last part is causal mask learning. First, a mask token is added as a learnable embedding vector to the spatial perception feature set. Second, the learnable positional encoding is injected into each element of the set to obtain a causal mask sequence. Third, causal mask sequence is added to the causal mask sequence and input into the causal mask learning module, where the multihead attention mechanism is employed to capture the interdependency between spatially neighboring pixels and center pixels. In addition, a multilayer perceptron is utilized to generate the causal mask vector for each target pixel within the HSI patch. The causal mask vector encodes the probability of a pixel belonging to the "causal region" (the area containing sufficient category discriminability information) or the "noncausal region" (the area containing heterogeneous pixels and spectral anomaly pixels). The causal mask vector corresponding to a target pixel $\mathbf{m}_i$ can be represented as follows:

$$\begin{cases} \hat{\boldsymbol{m}}_i = \left[\tilde{z}_i^{\text{mask}}; \tilde{z}_i^{\text{spa}}\right] + \boldsymbol{E}_{\text{pos}} \\ \tilde{\boldsymbol{m}}_i = \text{MHSA}\left(\text{LN}\left(\hat{\boldsymbol{m}}_i\right)\right) + \hat{\boldsymbol{m}}_i \\ \boldsymbol{m}_i = \text{MLP}\left(\text{LN}\left(\tilde{\boldsymbol{m}}_i\right)\right) \end{cases} \tag{2}$$

where $\tilde{z}_i^{\text{mask}}$ represents the learnable mask token, $\boldsymbol{E}_{\text{pos}}$ denotes the learnable position encode, LN$(\cdot)$ signifies the layer normalization operation, MLP$(\cdot)$ corresponds to the multilayer perceptron, and MHSA$(\cdot)$ stands for the multihead attention mechanism.

Finally, based on the causal mask vector, we obtain the causal mask matrix $\boldsymbol{M}$ using the argmax operation. Given that target pixels lacking category discriminability are likely to be assigned higher mask probabilities, the causal mask matrix enables us to effectively identify noncausal factors without category discriminability. To ensure the masking process can be optimized through backpropagation, we employ the Gumbel-Softmax to implement a smooth approximation of argmax [41], [42]

$$\begin{cases} \boldsymbol{M}_i = \text{Softmax}(\log(\boldsymbol{m}_i) + \mathbf{G}_i), & i = 1, \ldots, N \\ \mathbf{G}_i = -\log(-\log(\boldsymbol{\pi}_i)), & \boldsymbol{\pi}_i \sim U(0, 1) \end{cases} \tag{3}$$

where N represents the number of target pixels in HSI, $\sim$ represents the sampling operation, and $U(0, 1)$ represents a uniform distribution between 0 and 1.

2) Causal Intervention: Based on the causal mask matrix, we perform causal interventions on HSIs to obtain the counterfactual HSIs. Then, we maximize the correlation between these HSIs and counterfactual HSIs within the feature space, thus identifying the causal factors that remain stable under such interventions

$$\max_g \frac{1}{B} \sum_{b=1}^{B} \text{COR}\left(g_b(\boldsymbol{x}), g_b\left(\bar{\boldsymbol{x}}\right)\right) \tag{4}$$

where $g(\cdot)$ represents the feature extraction module, B refers to the dimension of features, and $g_b(\boldsymbol{x})$ and $g_b(\bar{\boldsymbol{x}})$ represents the bth dimension feature of the HSI $\boldsymbol{x}$ and counterfactual HSI $\bar{\boldsymbol{x}}$, respectively. COR$(\cdot)$ is used to measure the correlation between features before and after perturbation, where the cosine similarity is chosen. The "counterfactual" denotes a scenario that cannot be directly observed in the real world. We generate counterfactual HSIs by substituting pixels of low category discriminability in the HSIs with chosen center pixels from heterogeneous HSIs, thus emulating the disturbances caused by differing categories and spectrally anomalous pixels. In this way, the model can identify causal relationships between pixels and categories, thereby effectively reducing the information interference caused by factors that are noncausal related to the model's learning process.

The CIM intervenes in noncausal factors, consequently eliminating spurious statistical associations and enhancing the model's generalization performance. Furthermore, this intervention alters the model's dependency relationship, facilitating improved learning and inference of relationships between causal factors and semantic categories, which, in turn, boosts the model's performance.

E. Feature Extraction Module Based on Vision Transformer

In the feature extraction module, we first divide the hyperspectral sample into a spatial token sequence based on pixels and reuse the spectral feature extraction module of mask Transformer to capture the dependency relationship between HSI spectral bands. Then, the vision Transformer [43] is used to capture the spatial contextual information of HSI. In vision Transformer, a category token is added to capture global spatial information. Subsequently, we employ the multihead attention mechanism to capture contextual information within the spatial token sequence of HSIs, which results in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

extraction of discriminative features z

$$\begin{cases} z = \text{CONCAT}(\text{head}_1, \ldots, \text{head}_L)\boldsymbol{W}^O \\ \text{head}_l = \text{Softmax}\left(\dfrac{\boldsymbol{Q}_l \boldsymbol{K}_l^{\text{T}}}{B}\right)\boldsymbol{V}_l \end{cases} \tag{5}$$

where CONCAT$(\cdot)$ represents the concatenation operation, L is the number of attention heads, $\boldsymbol{W}^O$ corresponds to the learnable network parameters, and head$_l$ is the output of the lth attention head. $\boldsymbol{Q}_l$, $\boldsymbol{K}_l$, and $\boldsymbol{V}_l$ stand for the query vector, key vector, and value vector of single attention head, respectively, which can be acquired through linear mapping of original spectral features.

F. Independent Causal Constraints

This section focuses on optimizing the joint independence of causal factors, ensuring that the causal factors extracted from samples adhere to the independent causal mechanisms principle [38], thus minimizing the interference of redundant information on causal factors. The optimization objective can be expressed as follows:

$$\min_g \frac{1}{B(B-1)} \sum_{b \neq j} \text{COR}\left(g_b(\boldsymbol{x}), g_j(\bar{\boldsymbol{x}})\right) \tag{6}$$

where $g_j(\bar{\boldsymbol{x}})$ represents the jth dimension feature of the counterfactual HSI. Considering that the features between HSIs and counterfactual HSIs should possess the following attributes: high correlation in corresponding dimensions to ensure that their interference with noncausal factors remains constant; independence between different dimensions to minimize the redundancy of information across dimensions. As a result, the application of independent causal constraints can be expressed as follows:

$$L_{\text{I}} = \|\boldsymbol{C} - \boldsymbol{I}\|^2 \tag{7}$$

where $\boldsymbol{C}$ represents the correlation matrix of features between the counterfactual hyperspectral samples and original samples in each dimension and $\boldsymbol{I}$ represents the identity matrix. Minimizing the difference between $\boldsymbol{C}$ and $\boldsymbol{I}$ ensures the complete separation of causal and noncausal factors, as well as the clean and independent representation of causal factors. This satisfies the first two fundamental attributes of ideal causal factors.

G. Meta-Transfer Learning

First, we conduct the category sampling from the meta-training set to obtain a few-shot scenario training set containing C categories, with A samples per category. Next, R samples are randomly chosen from each category as support samples, while the remaining samples serve as query samples to construct few-shot scenario tasks. It is important to note that sample labels are reassigned in each scenario task according to the order of category sampling. This approach enables the network to induct general few-shot classification knowledge and fully generalize to few-shot classification scenarios involving heterogeneous pixels in the target domain.

Second, support and query samples are input into the feature extraction module. Under the guidance of the CIM and with the application of independent causal constraints, this module outputs support and query features that containing pure causal factors. After inputting the output features of the feature extraction module into ADAM [47], the domain discriminator will determine whether the feature belongs to the source domain or the target domain. Due to the setting of adversarial optimization goal, the feature extraction module will output domain-invariant features to deceive the domain discriminator. The domain-adaptation loss L_{d} can be expressed as follows:

$$L_{\text{d}} = \min_D \max_g E\left[\log D\left(g\left(\boldsymbol{x}^{\text{s}}\right)\right)\right] + E\left[\log\left(1 - D\left(g\left(\boldsymbol{x}^{\text{t}}\right)\right)\right)\right] \tag{8}$$

where $\boldsymbol{x}^{\text{s}}$ and $\boldsymbol{x}^{\text{t}}$ represent HSI in the source and target domains, respectively, $E[\cdot]$ denotes the expectation, and D represents the domain discriminator.

Then, the category prototypes are computed using the features of support samples. The $\boldsymbol{O}_c$, which represents prototype of category c, can be represented as follows:

$$\boldsymbol{O}_c = \frac{1}{R} \sum_{r=1}^{R} z_c^r \tag{9}$$

where z_c^r represents the support feature of category c. Next, the category predictions of query samples are obtained by measuring the distance between query feature and every category prototype in the feature space. Specifically, the probability that query feature z_q^Q belongs to category c can be expressed as follows:

$$P\left(\boldsymbol{Y}_q^Q = c | z_q^Q \in Q\right) = \frac{\exp\left(-d\left(z_q^Q, \boldsymbol{O}_c\right)\right)}{\sum_{c=1}^{C} \exp\left(-d\left(z_q^Q, \boldsymbol{O}_c\right)\right)} \tag{10}$$

where $\boldsymbol{Y}_q^Q$ represents the category label, Q represents the set of query feature, and $d(\cdot)$ is the Euclidean distance metric.

Finally, by minimizing few-shot classification loss, the model can conclude transferable meta-knowledge from few-shot scenario tasks to complete few-shot classification tasks in the target domain. The few-shot classification loss L_{fsl} can be expressed as follows:

$$L_{\text{fsl}} = \sum_{q=1}^{C \times (A-R)} L_e\left(\widehat{\boldsymbol{Y}}_q^Q, \boldsymbol{Y}_q^Q\right) \tag{11}$$

where $\widehat{\boldsymbol{Y}}_q^Q$ is the category prediction for query set and $L_e(\cdot)$ denotes the cross-entropy loss.

H. Causal Association Module

While causal intervention can remove noncausal factors, it cannot guarantee that the learned causal factors have enough discriminability for the categories using only supervised labels in few-shot scenario [44]. Therefore, we propose a causal correlation module that maximizes the mutual information between feature and category predictions, ensuring a strong causal association between feature and category semantics

$$\max_g H(Z, Y) = \iint P_{Z,Y}(z, y) \log \frac{P_{Z,Y}(z, y)}{P_Z(z) P_Y(y)} dz dy \tag{12}$$

where $P_{Z,Y}$ is the joint distribution of Z and Y, and P_Z and P_Y are their marginal distributions. Given the considerable challenge of quantifying marginal and joint distributions

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

in high-dimensional feature spaces, we establish a function mapping based on Donsker–Varadhan representation [45] of Kullback-Leibler (KL) divergence to represent mutual information

$$H(Z,Y) \geq H_{\Theta}(Z,Y) = \sup_{T:\,\Omega\to\mathbb{R}} E_{P_{Z,Y}}[T] - \log\left(E_{P_Z P_Y}\left[e^{T}\right]\right) \tag{13}$$

where $T : \Omega \to \mathbb{R}$ maps variables to real numbers, Θ represents a family of functions that can achieve the above mapping, and sup denotes the supremum. Since the expectation in the equation above cannot be directly computed, the estimation of mutual information is obtained by sampling from the data to estimate the expectation, and the supremum estimation $\widehat{H(Z,Y)}_M$ can be represented as follows:

$$\widehat{H(Z,Y)}_M = \sup_{\theta\in\Theta} E_{P^{(M)}_{Z,Y}}[T_\theta] - \log\left(E_{P^{(M)}_Z P^{(M)}_Y}[e^{T_\theta}]\right) \tag{14}$$

where M represents the number of sampled examples, T_θ represents the mutual information metric constructed using a neural network, and θ represents the network parameter.

In order to avoid computing integrals, Monte Carlo integration is used here. The causal association loss L_{C} can be expressed as follows:

$$L_{\mathrm{C}} = \sum_{m=1}^{M} T_\theta\left(z_m, \hat{\boldsymbol{Y}}^{Q}_{m}\right) - \log\left(\sum_{m=1}^{M} e^{T_\theta\left(z_m, \tilde{\boldsymbol{Y}}^{Q}_{m}\right)}\right) \tag{15}$$

where $(z_m, \hat{\boldsymbol{Y}}^{Q}_{m})$ represents the positive sample pairs sampled from the joint distribution of feature distribution Z and category prediction distribution Y, while $(z_m, \tilde{\boldsymbol{Y}}^{Q}_{m})$ represents the negative sample pairs sampled from the marginal distribution of Z and Y.

As merely concatenating features and category prediction vectors may not adequately capture the multimodal structures underlying complex data distributions, this article, inspired by Li et al. [46], employs multilinear mapping to embed $(z_m, \hat{\boldsymbol{Y}}^{Q}_{m})$ and $(z_m, \tilde{\boldsymbol{Y}}^{Q}_{m})$ into a reproducing kernel Hilbert space. This allows us to obtain joint distribution and marginal distribution information of features and category prediction vectors across different dimensions

$$\begin{cases} G_{\otimes}\left(z_m, \hat{\boldsymbol{Y}}^{Q}_{m}\right) = z_m \otimes \hat{\boldsymbol{Y}}^{Q}_{m} \\ G_{\otimes}\left(z_m, \tilde{\boldsymbol{Y}}^{Q}_{m}\right) = z_m \otimes \tilde{\boldsymbol{Y}}^{Q}_{m} \end{cases} \tag{16}$$

where $G_{\otimes}$ represents the multilinear mapping.

I. Total Loss

The total loss of both the source and target domains can be expressed as follows:

$$L_{\mathrm{total}} = L_{\mathrm{fsl}} + \lambda_1 L_{\mathrm{I}} + \lambda_2 L_{\mathrm{C}} + \lambda_3 L_{\mathrm{d}} \tag{17}$$

where λ_1–λ_3 are the loss weights.

III. Experiments

A. HSI Datasets

To fully validate the effectiveness of CMTL, the following four public HSI datasets were selected for experiments: Chikusei (CH), IP, Pavia University (PU), Salinas (SA), and Houston (HS).

The CH dataset was captured by the hyperspectral visible and near-infrared compact infrared imaging spectrometer (HVNIR-C) sensor in Chikusei, Japan, and contains 2517 × 2335 ground pixels, 128 bands, and 19 categories.

The IP dataset was captured by the airborne visible infrared imaging spectrometer (AVIRIS) sensor in IN, USA, and contains 145 × 145 ground pixels, 200 bands, and 16 categories.

The PU dataset was captured by the reflective optics system imaging spectrometer (ROSIS) sensor at Pavia University and contains 1096 × 715 ground pixels, 102 bands, and nine categories.

The SA dataset was captured by the AVIRIS sensor in Salinas, USA, and contains 512 × 127 ground pixels, 204 bands, and 16 categories.

The HS dataset was captured by the ITERS CASI-1500 566 sensor in Houston, TX, USA, where CASI stands for compact airborne spectrographic imager. HS dataset contains 349 × 1905 ground pixels, 144 bands, and 15 categories.

The types and numbers of ground objects in the HSI datasets used in the experiments are described in Table I.

B. Experimental Setting

During the experiments, CH was regarded as the source domain, while IP, PU, SA, and HS were treated as the target domain. Table I demonstrates that the categories in the source and target domains are different. All experiments were conducted on a computer equipped with a 3.80-GHz Intel Core I7-10700KF CPU, 32 GB of RAM, and an RTX 2080Ti GPU, and the experimental environment was PyTorch. The comparative methods include non-meta-learning methods: support vector machine (SVM) [48] and 3-dimensional CNN (3DCNN) [49]; meta-learning methods: DFSL [27] and siamese-prototype network (SPnet) [25]; and meta-transfer learning methods: DCFSL [32], Gia-CFSL [33], and GMTN [31].

To ensure a fair comparison, we followed the experimental settings of DCFSL and Gia-CFSL. For non-meta-learning methods, five labeled samples per category in the target domain were used for training. For meta-learning and meta-transfer learning methods, five labeled samples per category in the target domain and 200 labeled samples per category in the source domain were used for training. The maximum number of iterations is set to 1000, and Adam was used as the optimizer with a learning rate of 0.001. The patch size in CMTL is set to 9 × 9. The output dimension of DUM is 100, all hidden layer dimensions in the CIM and feature extraction module are set to 64, and the hidden layer dimension of discriminator in the ADAM is set to 128. In mask Transformer, the number of blocks is 5. Among them, four Transformer blocks are used to learn spectral features, and one is used to learn causal masks. In the feature extraction module, the number of Transformer blocks is 6. Among them, the four Transformer blocks used for learning spectral features are shared with mask Transformer, while the other two Transformer blocks are used to capture spatial dependencies between pixels in HSI. The parameter settings of each comparative method followed the original settings. To eliminate the influence of randomness, all experimental results were averaged over ten runs. Three evaluation metrics were chosen, including

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DESCRIPTION OF HSI DATASETS

Chikusei			Indian Pines			Pavia University			Salinas			Houston		
	Category	Number		Category	Number		Category	Number		Category	Number		Category	Number
C1	Water	2845	I1	Alfalfa	46	P1	Asphalt	6631	S1	Brocoli-green-weeds-1	2009	H1	Healthy grass	1251
C2	Bare-soil (school)	2859	I2	Corn-notill	1428	P2	Meadows	18649	S2	Brocoli-green-weeds-2	3726	H2	Stressed grass	1254
C3	Bare-soil (park)	286	I3	Corn-mintil	830	P3	Gravel	2099	S3	Fallow	1976	H3	Synthetic grass	697
C4	Bare-soil (farmland)	48525	I4	Corn	237	P4	Trees	3064	S4	Fallow-rough-plow	1394	H4	Trees	1244
C5	Natural plants	4297	I5	Grass-pasture	483	P5	Sheets	1345	S5	Fallow-smooth	2678	H5	Soil	1242
C6	Weeds in farmland	1108	I6	Grass-trees	730	P6	Bare-soil	5029	S6	Stubble	3959	H6	Water	325
C7	Forest	20516	I7	Grass-pasture-mowed	28	P7	Bitumen	1330	S7	Celery	3579	H7	Residential	1268
C8	Grass	6515	I8	Hay-windrowed	478	P8	Bricks	3682	S8	Grapes-untrained	11271	H8	Commercial	1244
C9	Rice field (grown)	13369	I9	Oats	20	P9	Shadow	947	S9	Soil-vinyard-develop	6203	H9	Road	1252
C10	Rice field (first stage)	1268	I10	Soybean-notill	972				S10	Corn-senesced-green-weeds	3278	H10	Highway	1227
C11	Row crops	5961	I11	Soybean-mintill	2455				S11	Lettuce-romaine-4wk	1068	H11	Railway	1235
C12	Plastic house	2193	I12	Soybean-clean	593				S12	Lettuce-romaine-5wk	1927	H12	Parking-lot-1	1233
C13	Manmade (non-dark)	1220	I13	Wheat	205				S13	Lettuce-romaine-6wk	916	H13	Parking-lot-2	469
C14	Manmade (dark)	7664	I14	Woods	1265				S14	Lettuce-romaine-7wk	1070	H14	Tennis court	428
C15	Manmade (black)	431	I15	Buildings-grass-trees-drives	386				S15	Vinyard-untrained	7268	H15	Running trac	660
C16	Manmade (red)	222	I16	Stone-steel-towers	93				S16	Vinyard-vertical-trellis	1807			
C17	Manmade grass	4040												
C18	Asphalt	801												
C19	Paved ground	145												
	Total	77592		Total	10249		Total	42776		Total	54129		Total	15029

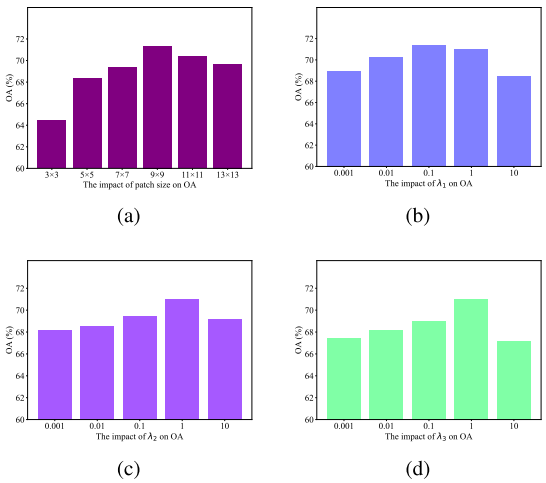


Fig. 4. OAs achieved by CMTL under different parameter settings (IP). (a) OA versus patch size. (b) OA versus λ_1. (c) OA versus λ_2. (d) OA versus λ_3.

overall accuracy (OA), average accuracy (AA), and kappa coefficient.

C. Parameter Analysis

We analyze the influence of different parameter settings on the performance of CMTL, including patch size p, loss weights λ_1–λ_3, and learning rate. We used the IP dataset as an example, with five labeled samples for each category, to analyze the impact of different parameter settings on CMTL performance.

1) Analysis of Patch Size: To analyze the impact of patch size on classification performance of CMTL, we set the patch size to $\{3 \times 3, 5 \times 5, 7 \times 7, 9 \times 9, 11 \times 11, 13 \times 13\}$. The variation of OA with patch size is shown in Fig. 4(a).

As the patch size increases, the classification accuracy of CMTL first increases and then decreases, and the best classification accuracy is achieved when the patch size is 9×9. This is because small patch sizes are difficult to fully utilize the rich spatial information of HSI, resulting in suboptimal performance. Utilizing a patch size that is excessively large can inadvertently incorporate noise information from the spatial vicinity into the model, potentially obstructing further improvements in the model's classification performance. Therefore, we set the patch size to 9×9 in all experiments.

2) Analysis of Loss Weight: To analyze the impact of loss weights on the classification performance of CMTL, we uniformly set λ_1–λ_3 to $\{0.001, 0.01, 0.1, 1, 10\}$. Fig. 4(b)–(d), respectively, shows the variation trend of OA with λ_1–λ_3 when the weights of other losses are set to 1, which show the following observations:

1) When λ_1 is small, it is difficult for the model to fully separate causal and noncausal factors. Large λ_1 makes model pay insufficient attention to few-shot learning task, which affects the classification performance of the model.
2) When λ_2 is small, it is difficult for the model to ensure a strong causal correlation between causal factors and category semantics, resulting in insufficient category discriminability information contained in features. When λ_2 is large, CMTL is easy to pay too much attention to a single task, which is not conducive to the overall optimization.
3) When λ_3 is small, CMTL cannot fully adapt the distribution of two domains. In contrast, it may lead to over-adaptation.

Note that for simplicity, the other datasets are set to the same parameters as IP, and we set $\lambda_1 = 0.1, \lambda_2 = 1$, and $\lambda_3 = 1$.

3) Analysis of Learning Rate: To analyze the impact of learning rate on classification performance of CMTL, we set the learning rate to $\{1e-5, 1e-4, 1e-3, 1e-2, 1e-1\}$, and the variation of OA with learning rate is shown in Fig. 5.

The model's classification accuracy shows an initial increase followed by a decrease as the learning rate increases. This phenomenon can be attributed to the fact that a smaller learning rate increases the risk of the model getting trapped

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

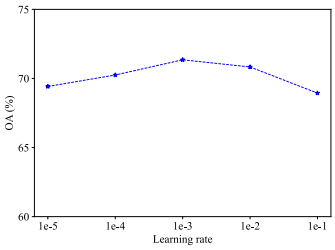


Fig. 5. OA achieved by CMTL under different learning rates (IP).

TABLE II
CLASSIFICATION PERFORMANCE ON IP (FIVE LABELED SAMPLES PER CATEGORY)

	SVM	3DCNN	DFSL	SPnet	DCFSL	Gia-CFSL	GMTN	CMTL
I1	51.22	**97.56**	92.68	80.49	87.80	91.60	56.10	95.12
I2	19.89	29.59	39.85	43.01	45.40	50.46	**53.76**	51.09
I3	47.88	35.88	56.85	61.58	57.21	44.88	48.24	**88.61**
I4	56.47	50.00	71.55	81.90	**84.91**	81.61	79.31	80.17
I5	82.01	47.49	64.64	72.80	**78.45**	70.76	**78.45**	65.90
I6	86.21	66.21	89.52	84.28	88.97	84.25	77.93	**94.35**
I7	95.65	**100**	**100**	**100**	**100**	97.10	**100**	**100**
I8	78.22	76.11	73.36	69.98	73.57	**91.12**	84.14	58.99
I9	**100**	86.67	**100**	**100**	**100**	99.26	93.33	**100**
I10	51.91	44.47	51.09	55.64	65.25	62.68	31.85	**76.32**
I11	48.94	59.02	52.28	60.53	59.14	66.54	**75.95**	58.98
I12	26.02	25.00	29.08	33.84	43.37	42.06	64.96	**71.94**
I13	91.50	98.00	98.00	97.00	99.00	97.11	99.00	**100**
I14	48.65	85.63	82.30	88.49	**90.16**	87.10	89.44	81.75
I15	20.73	38.06	54.07	52.49	62.47	68.74	65.88	**87.93**
I16	90.91	93.18	**100**	98.86	**100**	97.47	86.36	98.86
OA (%)	49.79	54.10	59.55	63.79	66.40	67.42	68.36	**71.35**
AA (%)	62.26	64.55	72.20	73.81	77.23	77.05	74.04	**81.88**
Kappa	43.70	48.23	54.43	59.03	62.08	63.10	64.03	**67.97**

TABLE III
CLASSIFICATION PERFORMANCE ON PU (FIVE LABELED SAMPLES PER CATEGORY)

	SVM	3DCNN	DFSL	SPnet	DCFSL	Gia-CFSL	GMTN	CMTL
P1	51.39	59.06	69.53	**89.86**	81.06	81.16	76.99	89.77
P2	67.62	72.14	84.46	83.63	87.74	89.58	**96.65**	92.96
P3	54.35	60.74	67.91	**68.39**	63.33	59.77	55.16	56.59
P4	69.30	73.46	76.72	82.64	92.56	**93.17**	70.32	92.87
P5	**100**	98.51	**100**	**100**	99.01	99.46	98.88	**100**
P6	49.26	51.91	48.09	53.42	74.58	**76.95**	76.89	64.89
P7	72.75	69.58	69.81	65.28	77.74	79.58	78.26	**87.02**
P8	68.70	43.05	80.61	83.00	62.42	74.36	74.15	**87.33**
P9	77.18	83.44	86.31	89.60	98.22	**98.56**	98.73	94.37
OA (%)	63.89	65.76	76.24	80.25	83.05	84.47	84.97	**86.97**
AA (%)	67.84	67.99	75.94	79.54	81.85	83.62	80.67	**85.09**
Kappa	54.06	55.90	68.59	73.88	77.46	79.71	79.99	**85.08**

in local optima, whereas a larger learning rate may cause the model to overshoot the optimal point during parameter updates, resulting in suboptimal accuracy. Therefore, we set the learning rate to $1e-3$ in all experiments.

D. Comparative Experiments

From the classification performance results shown in Tables II–V, it can be seen that the following hold.

1) Compared with non-meta-learning methods, meta-learning methods demonstrate higher classification accuracy. This can be attributed to their ability to induct transferable few-shot classification knowledge in the source domain through scenario training mechanisms and enhance target-domain few-shot classification accuracy via cross-domain knowledge transfer.
2) Meta-transfer methods outperform meta-learning methods in terms of classification performance. This is because meta-transfer methods can achieve more effective cross-domain knowledge transfer by reducing the distribution difference between two domains, which further improves the performance of few-shot classification models in the target domain.
3) CMTL achieves best classification performance in cross-domain few-shot HSI classification tasks. The advantages of CMTL can be ascribed to two aspects: on the one hand, by utilizing causal intervention and causal factor constraint, CMTL extracts pure causal factors, effectively suppressing the information interference of noncausal factors. On the other hand, CMTL employs a CAM to ensure a strong correlation between feature

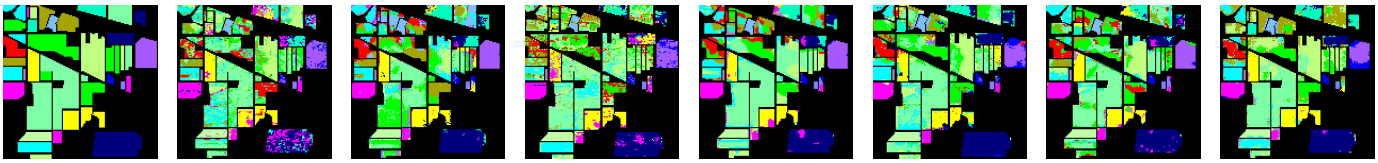
TABLE IV
CLASSIFICATION PERFORMANCE ON SA (FIVE LABELED SAMPLES PER CATEGORY)

	SVM	3DCNN	DFSL	SPnet	DCFSL	Gia-CFSL	GMTN	CMTL
S1	88.37	91.92	94.66	94.41	99.55	**99.88**	99.30	99.25
S2	78.85	79.84	99.00	98.31	**99.71**	99.34	97.83	99.52
S3	63.06	63.01	72.35	72.81	93.68	95.04	81.63	**99.95**
S4	77.68	81.14	93.38	93.02	**99.45**	96.99	94.53	87.40
S5	83.84	86.53	83.84	83.43	90.39	89.02	**97.42**	94.09
S6	99.97	99.97	99.37	99.52	99.27	99.52	99.67	**100**
S7	95.55	96.31	98.68	98.88	99.04	99.44	95.27	**99.52**
S8	46.09	52.46	68.60	68.26	72.61	80.89	81.34	**85.76**
S9	99.73	98.81	97.42	97.08	99.74	99.55	99.68	**99.77**
S10	69.84	78.95	76.54	79.38	84.51	76.47	94.99	**95.69**
S11	74.41	88.05	98.02	98.02	98.17	97.74	99.42	**99.53**
S12	98.13	97.55	98.49	98.07	**99.04**	98.39	92.72	96.20
S13	98.35	88.14	92.97	92.86	**98.97**	98.87	95.50	95.39
S14	85.54	96.34	98.78	98.40	97.99	95.69	95.77	**99.72**
S15	**88.92**	88.56	81.34	85.01	74.12	75.76	76.57	69.89
S16	72.81	72.09	88.12	86.51	90.62	94.08	94.67	**99.06**
OA (%)	78.72	81.18	86.22	86.65	88.53	89.95	90.57	**91.73**
AA (%)	82.57	84.98	90.10	90.25	93.55	93.54	93.53	**95.05**
Kappa	76.56	79.24	84.74	85.22	87.27	88.82	89.50	**90.79**

TABLE V
CLASSIFICATION PERFORMANCE ON HS (FIVE LABELED SAMPLES PER CATEGORY)

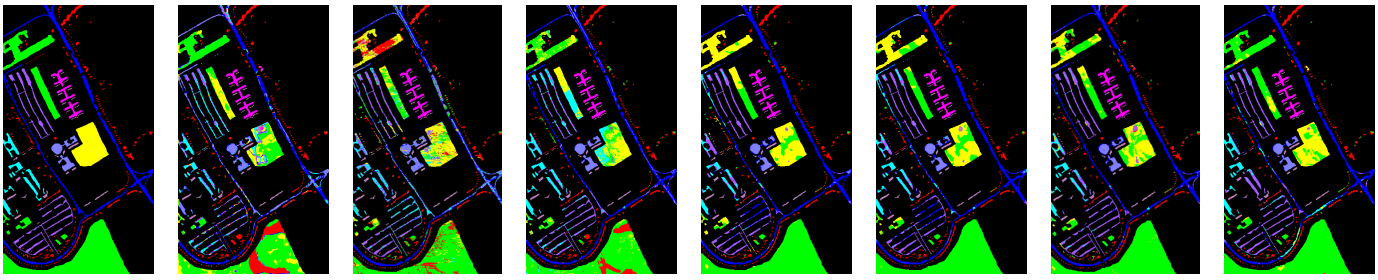
	SVM	3DCNN	DFSL	SPnet	DCFSL	Gia-CFSL	GMTN	CMTL
H1	85.55	96.47	94.62	90.53	82.58	84.75	96.14	**97.43**
H2	68.37	81.99	**93.11**	81.59	72.86	82.79	82.87	63.01
H3	99.86	78.90	99.28	95.09	90.75	84.39	99.42	**100**
H4	81.68	75.30	83.05	87.97	83.62	85.96	86.12	**93.46**
H5	98.63	97.33	94.02	**99.92**	93.94	92.00	92.40	99.19
H6	80.94	**89.69**	85.94	74.69	82.81	87.50	84.69	85.63
H7	21.38	55.66	79.49	77.44	73.00	54.87	**80.60**	70.23
H8	41.89	33.98	33.50	41.57	60.61	**65.62**	45.84	52.22
H9	31.92	73.06	50.60	69.85	63.83	71.53	**73.13**	63.03
H10	42.80	53.60	46.40	33.39	56.47	76.10	55.07	**80.03**
H11	**85.77**	64.96	54.96	64.55	57.40	50.00	62.28	81.06
H12	71.82	68.73	80.05	63.03	79.32	72.88	60.67	**93.08**
H13	67.90	66.81	71.34	76.08	**89.44**	84.91	86.85	81.68
H14	**100**	96.69	99.53	97.64	99.05	95.75	96.22	98.34
H15	99.69	69.92	79.08	**100**	88.09	94.35	99.39	89.31
OA (%)	67.81	71.53	59.55	63.79	75.43	76.31	77.20	**81.38**
AA (%)	71.88	73.54	72.20	73.81	78.25	78.89	80.11	**83.18**
Kappa	65.23	69.33	71.70	72.37	73.45	74.40	75.38	**79.88**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.



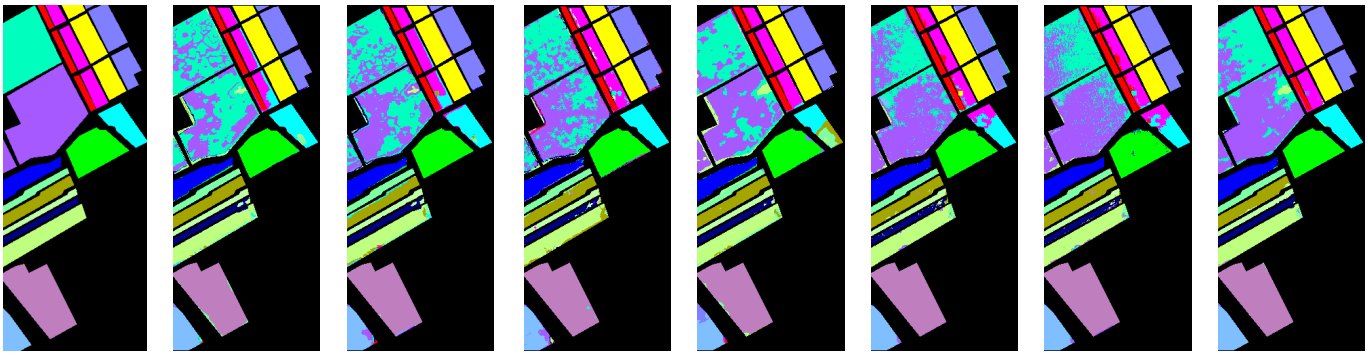
(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. Classification maps on IP. (a) Label map. (b) SVM. (c) 3DCCN. (d) DFSL. (e) SPnet. (f) DCFSL. (g) Gia-CFSL. (h) CMTL.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 7. Classification maps on PU. (a) Label map. (b) SVM. (c) 3DCCN. (d) DFSL. (e) SPnet. (f) DCFSL. (g) Gia-CFSL. (h) CMTL.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 8. Classification maps on SA. (a) Label map. (b) SVM. (c) 3DCCN. (d) DFSL. (e) SPnet. (f) DCFSL. (g) Gia-CFSL. (h) CMTL.

and category prediction. This results in features containing more category discriminative information, thereby improving the distinguishability between different categories.

E. Visual Analysis

Figs. 6–9 depict the classification accuracy maps and t-distributed stochastic neighbor embedding (t-SNE) maps of CMTL for the three groups of data. Fig. 10(a) shows a visualized image of an HSI patch, with yellow pixels being the same type of pixel as the center pixel, and black and green pixels being heterogeneous pixels. pixel β is the central pixel, while α and γ are two spectral anomaly pixels divided into noncausal regions. After the HSI patch is input into the causal mask network, it can output the causal mask shown in Fig. 10(b). The black part in the causal mask represents the reserved causal area, while the gray part represents the noncausal area that can be replaced. By replacing pixels in noncausal regions with randomly heterogeneous pixels, the counterfactual HSI shown in Fig. 10(c) can be obtained. We can observe that the following hold.

1) In Figs. 6–8, CMTL exhibits the best classification accuracy with the clearest boundaries and the highest intracategory consistency. This is because CMTL effectively separates causal factors and noncausal factors by utilizing causal intervention, reducing the impact of heterogeneous category pixels and spectral anomalies on discriminative features in the spatial neighborhood.
2) In Fig. 9, CMTL performs well in the t-SNE map, showing high intracategory consistency and intercategory separation. This is because the CAM maximizes the mutual information between features and category predictions, making features contain as much category-related information and as little heterogeneous information as possible. This not only promotes the aggregation of samples of the same category in the feature space but also ensures the separation of samples of different categories.
3) On the one hand, it can be seen from Fig. 10(a) and (b) that the causal mask effectively identifies the right region where heterogeneous pixels are located. On the other hand, Fig. 10(d) and (e) shows the spectral response of pixels α, β, and γ in various bands. When combined with Fig. 10(a) and (b), it can be observed that pixel α and pixel γ, which categorized into noncausal regions,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

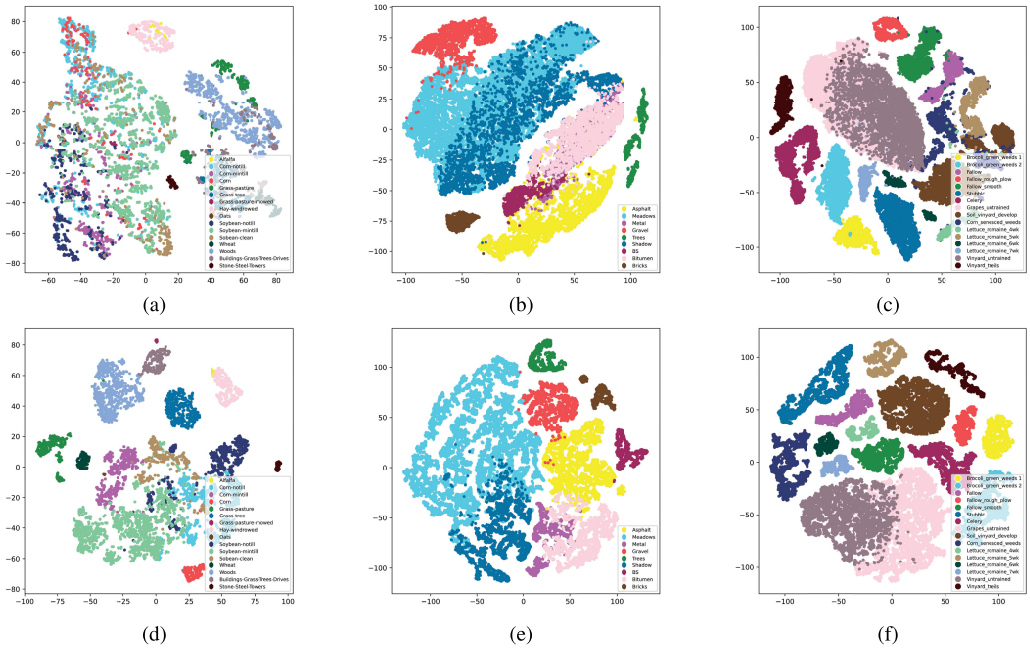


Fig. 9. t-SNE of HSI features. (a) Original IP features. (b) Original PU features. (c) Original SA features. (d) Features learned by CMTL on IP. (e) Features learned by CMTL on PU. (f) Features learned by CMTL on SA.

exhibit different spectral variation trends compared with the center pixel β. This indicates that causal masks can effectively identify spectral anomaly pixels.

F. Ablation Study

We performed ablation studies on ADAM, CIM, and CAM of CMTL to verify their effectiveness. Tables VI–VIII show the impact of different techniques on classification performance of CMTL.

It can be observed from Tables VI–VIII that the following hold.

1) Compared with CMTL-A, CMTL-B shows higher classification accuracy. This improvement can be attributed to the use of ADAM, which reduces the distribution difference between domains and, thereby, promotes cross-domain knowledge transfer.
2) Compared with CMTL-B, the classification accuracy of CMTL-C is higher across all four datasets, with the increases of 2.07% (PU), 1.72% (SA), 2.05% (IP), and 2.44% (HS), respectively. This is because CIM can suppress the interference of noncausal factors and help the model obtain pure causal factors. Consequently, the false statistical association between causal factors and category semantics is eliminated, improving the model's generalization performance.
3) Compared with CMTL-C, CMTL achieved the accuracy improvements of 1.35% (PU), 1.08% (SA), 1.73% (IP), and 2.73% (HS) on the four datasets, respectively. This improvement is due to the CAM, which maximizes the mutual information between HSI features and category predictions to ensure a strong causal correlation between causal factors and category semantics. As a result, the features contain more category discriminative information.

TABLE VI
ABLATION EXPERIMENTS

		CMTL-A	CMTL-B	CMTL-C	CMTL
Model Description	ADAM	×	✓	✓	✓
	CIM	×	×	✓	✓
	CAM	×	×	×	✓

TABLE VII
IMPACT OF DIFFERENT TECHNIQUES ON OA (%)

Dataset	CMTL-A	CMTL-B	CMTL-C	CMTL
PU	82.14	83.55	85.62	86.97
SA	87.86	88.93	90.65	91.73
IP	66.75	67.57	69.62	71.35
HS	72.65	76.21	78.65	81.38

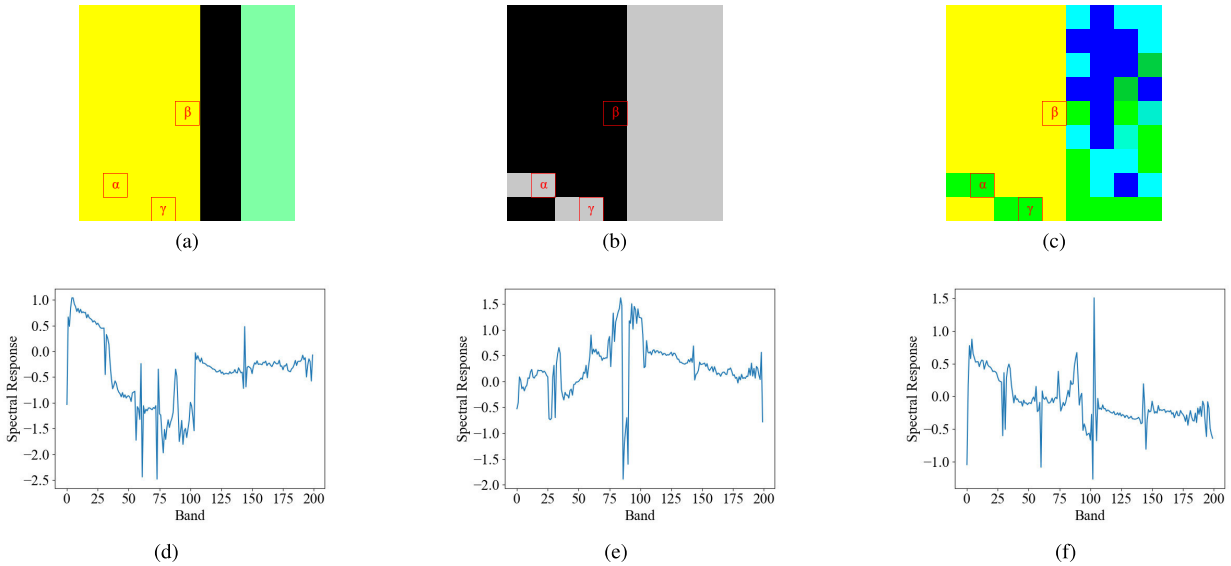
G. Adaptability Experiments on Different Numbers of Labeled Data

We set the number of labeled samples for model training to 1–5, respectively. We then compared the OAs of CMTL and comparative methods on the PU, IP, SA, and HS datasets under these settings. The results of these comparisons are presented in Fig. 11.

As can be seen from Fig. 11, the following hold.

1) As the quantity of labeled samples increases, there is a corresponding upward trend in the OA across various methods. This phenomenon can be attributed to the fact that an increased number of training samples results in a wider coverage of feature distributions. Consequently, the training set is better equipped to accurately reflect the actual distribution of diverse categories. This improved representation enables the model to learn the features of different categories with greater precision, thereby enhancing its predictive performance on test samples.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

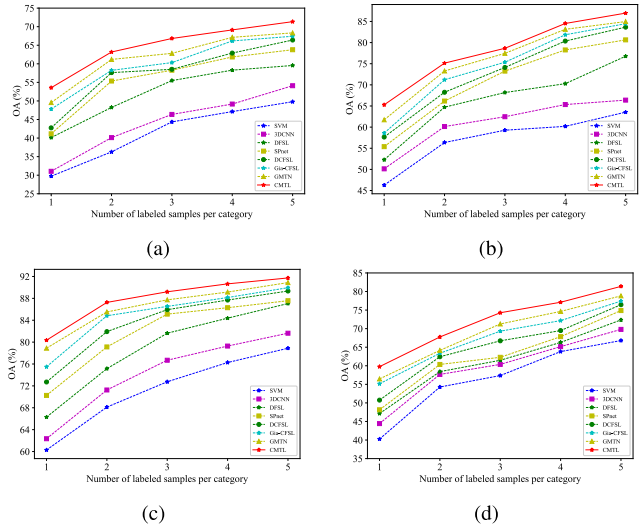


(a) (b) (c) (d) (e) (f)

Fig. 10. Visual analysis of causal mask. (a) HSI. (b) Causal mask. (c) Counterfactual HSI. (d) Spectral curve of pixel α. (e) Spectral curve of pixel β. (f) Spectral curve of pixel γ.

TABLE VIII
COMPUTATIONAL COST ON EACH DATASET WITH DIFFERENT METHODS

	Methods	SVM	3DCCN	DFSL	SPnet	DCFSL	Gia-CFSL	GMTN	CMTL
PU	Training time (s)	0.28	165.31	1423.12	1351.32	3817.35	4318.26	1398.55	1214.23
	Testing time (s)	1.93	6.57	7.13	7.28	8.36	5.52	5.17	4.82
	FLOPs (M)	-	15.63	41.13	43.13	46.19	9.86	8.97	8.32
	Parameters (M)	-	0.34	0.35	0.35	4.26	0.31	0.78	0.28
SA	Training time (s)	0.31	175.27	2314.35	1854.67	5145.37	6814.41	1515.34	1358.36
	Testing time (s)	2.36	7.94	8.13	8.25	9.53	7.62	7.28	7.13
	FLOPs (M)	-	17.80	41.13	43.13	47.01	9.86	8.97	8.32
	Parameters (M)	-	0.34	0.35	0.35	4.26	0.31	0.78	0.28
IP	Training time (s)	0.26	189.64	2145.95	2014.68	4863.41	6245.38	1683.6	1324.45
	Testing time (s)	1.57	2.36	4.82	4.13	4.82	1.63	1.46	1.32
	FLOPs (M)	-	17.80	41.13	43.13	46.98	9.86	8.97	8.32
	Parameters (M)	-	0.34	0.35	0.35	4.26	0.31	0.78	0.28
HS	Training time (s)	0.27	175.32	2013.26	1945.34	4632.54	6182.13	1597.25	1372.36
	Testing time (s)	1.64	3.13	4.86	5.26	5.42	1.93	1.87	1.56
	FLOPs (M)	-	17.70	41.13	43.13	46.98	9.86	8.97	8.32
	Parameters (M)	-	0.34	0.35	0.35	4.26	0.31	0.78	0.28



(a) (b) (c) (d)

Fig. 11. OAs (%) of different methods with different numbers of labeled samples per class. (a) IP. (b) PU. (c) SA. (d) HS.

2) CMTL delivered the highest classification accuracy across all datasets, demonstrating strong adaptability to varying quantities of labeled samples. This can be attributed to the fact that sample selection bias within few-shot scenarios can easily introduce noncausal factors into HSI, consequently diminishing the model's accuracy. However, CMTL effectively mitigates this issue through the separation of noncausal factors. This further proves that CMTL has significant advantages in few-shot learning tasks and can effectively use limited labeled samples to provide accurate classification results.

H. Computational Cost

In this section, we present the computational cost, which includes the computational time, floating-point operations (FLOPs), and the number of network parameters for different methods. The detailed results are summarized in Table VIII.

As can be seen from Tables II–IV and VIII, the following hold.

1) Compared with SVM, deep-learning methods require more computational cost, as they need to be optimized through backpropagation. However, this additional investment corresponds to a significant improvement in performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

2) Compared with 3DCNN, meta-learning methods need to encapsulate knowledge from the source domain, resulting in some time cost. However, because of their full utilization of source-domain data, these methods also achieve superior classification performance.
3) Meta-transfer methods, such as Gia-CFSL, GMTN, and CMTL, have higher computational costs than SPnet and DCFSL, because they require additional computations to adjust the distribution between domains.
4) CMTL not only achieves the highest classification accuracy but also maintains an acceptable computational cost. This indicates its effectiveness in addressing few-shot HSI classification task.

IV. Conclusion

The CMTL addresses the limitations of the existing meta-transfer learning methods in few-shot scenarios. The existing methods primarily model the statistical dependency between data and labels, which may lead to restricted generalization performance due to the few-shot nature of the scenarios. To mitigate this, we have incorporated causal learning into the framework of meta-transfer learning, which captures the causal association between data and category semantics to enable more generalizable few-shot learning. CMTL offers the following advantages: 1) eliminating false statistical associations between noncausal factors and category semantics through the causal intervention to improve model generalization performance; 2) constraining the joint independence of causal factors to suppress redundant information interference; and 3) maximizing the mutual information between features and category predictions to fully explore the potential causal associations between causal factors and category semantics. Experimental results demonstrate that CMTL achieves state-of-the-art cross-domain few-shot HSI performance on three public HSI datasets. At present, this article only explores the causal element transfer learning in a single-source-domain scenario. In the future work, we will try to extend CMTL to a multisource-domain scenario to further improve the generalization performance by using rich information of multiple source domains.

References

[1] C. Zhao, B. Qin, S. Feng, W. Zhu, L. Zhang, and J. Ren, "An unsupervised domain adaptation method towards multi-level features and decision boundaries for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546216.

[2] X. Shang, M. Song, and C.-I. Chang, "Multispatial filtering module cascaded system for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510521.

[3] W. Zhu, C. Zhao, S. Feng, and B. Qin, "Multiscale short and long range graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535815.

[4] J. Liu et al., "Estimating the forage neutral detergent fiber content of Alpine grassland in the Tibetan Plateau using hyperspectral data and machine learning algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4405017.

[5] C. Zhao, W. Zhu, and S. Feng, "Superpixel guided deformable convolution network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3838–3851, 2022.

[6] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[7] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[8] H. Liu, W. Li, X.-G. Xia, M. Zhang, C.-Z. Gao, and R. Tao, "Central attention network for hyperspectral imagery classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Mar. 10, 2022, doi: 10.1109/TNNLS.2022.3155114.

[9] Y. Ding et al., "Self-supervised locality preserving low-pass graph convolutional embedding for large-scale hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536016.

[10] Y. Ding et al., "Multi-scale receptive fields: Graph attention neural network for hyperspectral image classification," *Exp. Syst. Appl.*, vol. 223, Aug. 2023, Art. no. 119858, doi: 10.1016/j.eswa.2023.119858.

[11] Y. Ding et al., "Unsupervised self-correlated learning smoothy enhanced locality preserving graph convolution embedding clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536716.

[12] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[13] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[14] Y. Ding, X. Zhao, Z. Zhang, W. Cai, N. Yang, and Y. Zhan, "Semi-supervised locality preserving dense graph neural network with ARMA filters and context-aware learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511812.

[15] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.

[16] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[17] X. He, Y. Chen, and P. Ghamisi, "Heterogeneous transfer learning for hyperspectral image classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3246–3263, May 2020.

[18] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[19] R. Hang, X. Qian, and Q. Liu, "Cross-modality contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532812.

[20] Y. Qu, R. K. Baghbaderani, W. Li, L. Gao, Y. Zhang, and H. Qi, "Physically constrained transfer learning through shared abundance space for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10455–10472, Dec. 2021.

[21] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural. Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Sep. 2021.

[22] W. Wang, Z. Shen, D. Li, P. Zhong, and Y. Chen, "Probability-based graph embedding cross-domain and class discriminative feature learning for domain adaptation," *IEEE Trans. Image Process.*, vol. 32, pp. 72–87, 2023.

[23] M. Zhang, X. Zhao, W. Li, Y. Zhang, R. Tao, and Q. Du, "Cross-scene joint classification of multisource data with multilevel domain adaption network," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Apr. 6, 2023, doi: 10.1109/TNNLS.2023.3262599.

[24] T. Hospedales, A. Antoniou, P. Micaelli, and A. Storkey, "Meta-learning in neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5149–5169, Sep. 2022.

[25] G. Cheng et al., "SPNet: Siamese-prototype network for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5608011.

[26] J. Bai et al., "Few-shot hyperspectral image classification based on adaptive subspaces and feature transformation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523917.

[27] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.

[28] K. Gao, B. Liu, X. Yu, J. Qin, P. Zhang, and X. Tan, "Deep relation network for hyperspectral image few-shot classification," *Remote Sens.*, vol. 12, no. 6, p. 923, Mar. 2020.

[29] F. Sung, Y. Yang, L. Zhang, T. Xiang, P. H. S. Torr, and T. M. Hospedales, "Learning to compare: Relation network for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1199–1208.

[30] X. Zuo, X. Yu, B. Liu, P. Zhang, and X. Tan, "FSL-EGNN: Edge-labeling graph neural network for hyperspectral image few-shot classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526518.

[31] H. Wang, X. Wang, and Y. Cheng, "Graph meta transfer network for heterogeneous few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501112, doi: 10.1109/TGRS.2022.3233591.

[32] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[33] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 30, 2022, doi: 10.1109/TNNLS.2022.3185795.

[34] Y. Wang et al., "Heterogeneous few-shot learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[35] W. Wang, F. Liu, J. Liu, and L. Xiao, "Cross-domain few-shot hyperspectral image classification with class-wise attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502418.

[36] D. Zhang, H. Zhang, J. Tang, X. Hua, and Q. Sun, "Causal intervention for weakly-supervised semantic segmentation," in *Proc. NIPS*, Dec. 2020, pp. 655–666.

[37] F. Lv et al., "Causality inspired representation learning for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8036–8046.

[38] B. Schoelkopf, D. Janzin, J. Peters, E. Sgouritsa, K. Zhang, and J. Mooij, "On causal and anticausal learning," in *Proc. Int. Conf. Mach. Learn.*, Edinburgh, U.K., Jun. 2012, pp. 1–8.

[39] J. Jiang, J. Ma, C. Chen, Z. Wang, Z. Cai, and L. Wang, "SuperPCA: A superpixelwise PCA approach for unsupervised feature extraction of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4581–4593, Aug. 2018.

[40] H. Zhang, L. Xiao, X. Cao, and H. Foroosh, "Multiple adverse weather conditions adaptation for object detection via causal intervention," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, Apr. 12, 2022, doi: 10.1109/TPAMI.2022.3166765.

[41] I. A. M. Huijben, W. Kool, M. B. Paulus, and R. J. G. van Sloun, "A review of the Gumbel-max trick and its extensions for discrete stochasticity in machine learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 1353–1371, Feb. 2023.

[42] E. Jang, S. Gu, and B. Poole, "Categorical reparameterization with Gumbel-softmax," in *Proc. Int. Conf. Learn. Represent.*, Toulon, France, Apr. 2017, pp. 1–13.

[43] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[44] C. Xu et al., "PatchMix augmentation to identify causal features in few-shot learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 7639–7653, Jun. 2023, doi: 10.1109/TPAMI.2022.3223784.

[45] V. Kostina and E. Tuncel, "Successive refinement of abstract sources," *IEEE Trans. Inf. Theory*, vol. 65, no. 10, pp. 6385–6398, Oct. 2019.

[46] S. Li et al., "Domain conditioned adaptation network," in *Proc. AAAI Conf. Artif. Intell.*, New York, NY, USA, Feb. 2020, pp. 11386–11393.

[47] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2096–2030, 2016.

[48] F. Melgani and L. Bruzzone, "Support vector machines for classification of hyperspectral remote-sensing images," in *Dig. Int. Geosci. Remote Sens. Symp.*, Toronto, ONT, Canada, Jun. 2002, pp. 506–508.

[49] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.



Yuhu Cheng (Member, IEEE) received the Ph.D. degree in control science and technology from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2005.

He is currently a Professor with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, China. His main research interests include machine learning and intelligent system.



Wei Zhang received the B.E. degree in automation from Nanton University, Nanton, China, in 2020. He is currently pursuing the M.E. degree with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, China.

His research interests include meta-learning and transfer learning.



Haoyu Wang (Member, IEEE) received the M.S. and Ph.D. degrees from the China University of Mining and Technology, Xuzhou, China, in 2017 and 2021, respectively.

He is currently a Post-Doctoral Researcher with the School of Information and Control Engineering, China University of Mining and Technology. His main research interests include hyperspectral image analysis, meta-learning, and transfer learning.



Xuesong Wang (Member, IEEE) received the Ph.D. degree in control science and technology from the China University of Mining and Technology, Xuzhou, China, in 2002.

She is currently the Dean of the Engineering Research Center of Intelligent Control for Underground Space, Ministry of Education, the Dean of the Xuzhou Key Laboratory of Artificial Intelligence and Big Data, and a Professor with the School of Information and Control Engineering, China University of Mining and Technology. Her main research interests include reinforcement learning, machine learning, and pattern recognition.

Dr. Wang is an Associate Editor of the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, the IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS, the IEEE SIGNAL PROCESSING LETTERS, the *International Journal of Machine Learning and Cybernetics*, *Acta Automatica Sinica*, and *Acta Electronica Sinica*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 05:16:14 UTC from IEEE Xplore. Restrictions apply.